2.1.6 Cartesian Products

Definition 7: The **ordered n-tuple** $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, ..., and a_n as its nth element.

- We say that two ordered n-tuples are equal if and only if each corresponding pair of their elements is equal.
- In other words, $(a_1, a_2, \ldots, a_n) = (b_1, b_2, \ldots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2, \ldots, n$.
- Ordered 2-tuples are called **ordered pairs**.

Definition 8: Let A and B be sets. The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. Hence, $A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$.

Example 17: What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: $A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$ □

Example 18: Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$, where A and B are as in Example 17.

Solution: $B \times A = \{(a,1), (a,2), (b,1), (b,2), (c,1), (c,2)\}$

This is not equal to $A \times B$, which was found in Example 17. □

Definition 9: The Cartesian product of the sets $A_1, A_2, \ldots, A_n$, denoted by $A_1 \times A_2 \times \ldots \times A_n$, is the set of ordered n-tuples $(a_1, \ldots, a_n)$, where a_i belongs to A_i for $i = 1, 2, \ldots, n$. In other words,

$$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \ldots, n\}.$$

Example 19: What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?